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RESEARCH LABS

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PROJECT ON

Heart Disease Prediction Using Machine Learning Algorithms

About me

About me Background

- I am a 4th-year B.Tech student pursuing Computer Science Engineering.

Why you want to learn Data Science

- I want to learn Data Science to develop strong skills in data analysis, visualization, and problem-solving using real-world data. As a computer science student, I am interested in data-driven decision-making and aim to pursue a career in data-related roles.

Any work experience

- Fresher with hands-on experience in Excel, EDA , PowerBI and SQL-based data analysis projects, with a strong interest in data analytics roles.

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Business Problem

- Heart disease is one of the leading causes of death worldwide.
- Early prediction helps identify high-risk patients before serious complications occur.
- Manual risk assessment using multiple health factors can be complex and time-consuming.
- A machine learning model can assist healthcare professionals in making faster and more accurate predictions.
- Early diagnosis supports timely treatment and improves patient outcomes.

Problem Statement

- Predict whether a patient is likely to develop **Coronary Heart Disease (CHD)** within the next **10 years**.
- Use the **Framingham Heart Study** dataset for prediction.
- Preprocess the data by handling missing values, duplicates, and outliers.
- Perform Exploratory Data Analysis (EDA) to understand the dataset.
- Train and compare three machine learning models.
- Evaluate models using **Accuracy, Precision, Recall, and F1-Score**.
- Select the best-performing model for heart disease prediction.

Objectives

- Predict the likelihood of **Coronary Heart Disease (CHD)** within the next 10 years using machine learning.
- Analyze and understand the Framingham Heart Study dataset.
- Handle missing values and preprocess the dataset for model training.
- Perform Exploratory Data Analysis (EDA) to identify patterns and relationships.
- Train and compare **Logistic Regression, Decision Tree, and Random Forest** models.
- Evaluate model performance using **Accuracy, Precision, Recall, and F1-Score**.
- Identify the best-performing model for heart disease prediction.

CRISP-ML(Q) Methodology

Business Understanding

Defined the problem of predicting heart disease risk.

Data Understanding

Explored the dataset and analyzed missing values, duplicates, and data types.

Data Preparation

Handled missing values, detected outliers, scaled features, and selected important features.

Model Building

Trained Logistic Regression, Decision Tree, and Random Forest models.

Evaluation

Compared models using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Deployment

Saved the best-performing model as a **Pickle (.pkl)** file for future predictions.

Dataset Overview

Dataset Information

Dataset: Framingham Heart Study Dataset

Domain: Healthcare

Prediction Type: Binary Classification

Target Variable: TenYearCHD

0 → No Heart Disease

1 → Heart Disease

Dataset Statistics

Total Records: 4,238

Total Features: 16 (including target variable)

Data Types: 9 Float, 7 Integer

male	age	education	currentSmoker	cigsPerDay	BPMeds	prevalentStroke	prevalentHyp	diabetes	totChol	sysBP	diaBP	BMI	heartRate	glucose	TenYearCHD
1	39	4.0	0	0.0	0.0	0	0	0	195.0	106.0	70.0	26.97	80.0	77.0	0
0	46	2.0	0	0.0	0.0	0	0	0	250.0	121.0	81.0	28.73	95.0	76.0	0
1	48	1.0	1	20.0	0.0	0	0	0	245.0	127.5	80.0	25.34	75.0	70.0	0
0	61	3.0	1	30.0	0.0	0	1	0	225.0	150.0	95.0	28.58	65.0	103.0	1
0	46	3.0	1	23.0	0.0	0	0	0	285.0	130.0	84.0	23.10	85.0	85.0	0

Data Understanding

- Explored the dataset using `df.info()`, `df.describe()`, and `df.isnull().sum()`.
- Verified the dataset structure, data types, and target variable.
- Identified missing values in different features.
- Analyzed the statistical summary of numerical features.
- Used these insights to guide the data preprocessing process.

Key Observations

- **Total Records:** 4,238
- **Total Features:** 16 (including the target variable)
- **Target Variable:** **TenYearCHD** (Binary Classification)
- **Data Types:** 9 Float, 7 Integer
- **Missing Values:** Found in Education, CigsPerDay, BPMeds, Total Cholesterol, BMI, Heart Rate, and Glucose.
- **No missing values** in the target variable (TenYearCHD).

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4238 entries, 0 to 4237
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   male                  4238 non-null   int64
1   age                   4238 non-null   int64
2   education             4133 non-null   float64
3   currentSmoker         4238 non-null   int64
4   cigsPerDay            4209 non-null   float64
5   BPMeds                4185 non-null   float64
6   prevalentStroke       4238 non-null   int64
7   prevalentHyp          4238 non-null   int64
8   diabetes              4238 non-null   int64
9   totChol               4188 non-null   float64
10  sysBP                 4238 non-null   float64
11  diaBP                 4238 non-null   float64
12  BMI                   4219 non-null   float64
13  heartRate             4237 non-null   float64
14  glucose               3850 non-null   float64
15  TenYearCHD            4238 non-null   int64
dtypes: float64(9), int64(7)
memory usage: 529.9 KB
```

Data Preprocessing

Steps Performed

- Checked the dataset for missing values.
- Compared **Mean, Median, and KNN Imputation** techniques to handle missing data.
- Selected **Median Imputation** for the final model as it is less affected by outliers.
- Checked for duplicate records.
- Detected potential outliers using **Box Plots**.
- Applied **StandardScaler** to normalize numerical features before model training.

Key Outcomes

- Missing values were successfully handled.
- No duplicate records were found.
- Numerical features were standardized to improve model performance.
- The dataset was cleaned and prepared for machine learning.

	Missing Values	Percentage
education	105	2.48
cigsPerDay	29	0.68
BPMeds	53	1.25
totChol	50	1.18
BMI	19	0.45
heartRate	1	0.02
glucose	388	9.16

```
Missing values after Mean Imputation:
male      0
age       0
education 0
currentSmoker 0
cigsPerDay 0
BPMeds    0
prevalentStroke 0
prevalentHyp 0
diabetes  0
totChol   0
sysBP     0
diaBP     0
BMI       0
heartRate 0
glucose   0
TenYearCHD 0
dtype: int64
```

```
Missing values after Median Imputation:
male      0
age       0
education 0
currentSmoker 0
cigsPerDay 0
BPMeds    0
prevalentStroke 0
prevalentHyp 0
diabetes  0
totChol   0
sysBP     0
diaBP     0
BMI       0
heartRate 0
glucose   0
TenYearCHD 0
dtype: int64
```

```
Missing values after KNN Imputation:
male      0
age       0
education 0
currentSmoker 0
cigsPerDay 0
BPMeds    0
prevalentStroke 0
prevalentHyp 0
diabetes  0
totChol   0
sysBP     0
diaBP     0
BMI       0
heartRate 0
glucose   0
TenYearCHD 0
dtype: int64
```

Feature Scaling

Technique Used

Applied **StandardScaler** to numerical features.
Standardized features before model training.

Purpose

Bring all numerical features to a common scale.
Prevent features with larger values from dominating the model.
Improve the performance of machine learning algorithms.

Outcome

Features were standardized with a mean of **0** and a standard deviation of **1**.
Scaled data was used for training and testing.

Feature Scaling

```
}]:  
from sklearn.preprocessing import StandardScaler  
  
scaler = StandardScaler()  
  
X_train = scaler.fit_transform(X_train)  
  
X_test = scaler.transform(X_test)
```

Exploratory Data Analysis (EDA)

Analysis Performed

- Analyzed the distribution of numerical features.
- Identified relationships between different variables.
- Visualized feature distributions using histograms.
- Examined feature correlations using a heatmap.
- Detected potential outliers using box plots.

Key Insights

- Most numerical features showed a normal or slightly skewed distribution.
- Some features contained outliers, identified using box plots.
- Certain features showed stronger correlations with the target variable than others.
- EDA provided valuable insights for data preprocessing and model building.

Feature Selection

Technique Used

Applied **SelectKBest** for feature selection.

Used the **ANOVA F-test (f_classif)** to evaluate the relationship between each feature and the target variable.

Selected the **top 8 important features** for model training.

Purpose

Identify the most relevant features.

Remove less important or irrelevant features.

Reduce model complexity.

Improve model performance and efficiency.

Outcome

Features were ranked based on their ANOVA F-scores.

The top 8 features were selected for building the machine learning models.

Model Building

Machine Learning Models Used

Logistic Regression

- Used as the baseline classification model.
- Simple, fast, and interpretable.

Decision Tree

- Classifies data using decision rules.
- Easy to understand and visualize.

Random Forest

- Ensemble learning algorithm that combines multiple decision trees.
- Reduces overfitting and improves prediction accuracy.

Training Process

- Split the dataset into **80% Training** and **20% Testing**.
- Trained all three models using the training dataset.
- Compared model performance on the testing dataset.

Model Evaluation & Results

Evaluation Metrics

- **Accuracy** – Measures the overall correctness of the model.
- **Precision** – Measures how many predicted positive cases are actually positive.
- **Recall** – Measures how many actual positive cases are correctly identified.
- **F1-Score** – Harmonic mean of Precision and Recall.

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.861	0.714	0.081	0.145
2	Random Forest	0.855	0.529	0.073	0.128
1	Decision Tree	0.774	0.250	0.274	0.262

Conclusion

Project Summary

- Successfully developed a machine learning model for **Heart Disease Prediction**.
- Performed data preprocessing, including missing value handling and feature scaling.
- Conducted Exploratory Data Analysis (EDA) to understand the dataset.
- Trained and compared **Logistic Regression, Decision Tree, and Random Forest** models.
- Evaluated models using **Accuracy, Precision, Recall, and F1-Score**.
- **Logistic Regression** achieved the best overall performance with the highest accuracy (**86.1%**) and was selected as the final model.

Future Scope

- Perform hyperparameter tuning to improve model performance.
- Train on larger and more diverse healthcare datasets.
- Deploy the model as a web application using Flask or Streamlit.
- Integrate the model into clinical decision support systems.

Real-World Applications

- **Hospitals & Clinics:** Assist doctors in identifying patients at high risk of heart disease for early intervention.
- **Preventive Healthcare:** Help detect high-risk individuals before severe cardiac events occur.
- **Health Monitoring Apps:** Integrate with mobile or wearable health applications to provide personalized risk assessments.
- **Clinical Decision Support:** Support healthcare professionals by providing an additional prediction tool during diagnosis.
- **Health Screening Programs:** Help prioritize patients who require further medical tests and monitoring.
- **Healthcare Research:** Analyze patient data to study risk factors and improve preventive healthcare strategies.

THANK
YOU

